


Inspiring Excellence

# CSE422: Artificial Intelligence Project Report

## Bank Marketing Campaign Prediction

Section: 18 Group: 17

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# 1. Introduction

Banks mostly run its marketing campaigns by running  outbound phone calls to convencse clients to subscribe to term deposits. Contacting every single customer from the database is time-consuming and very expensive. However, most calls do not result in a sale. Our project addresses that inefficiency by building a machine learning model that can predict before a call is made, whether a given potential client will subscribe to a term deposit or not.

We used a given  Bank Marketing dataset which contains 41188 records describing client demographics, past campaign contact history and broader economic indicators at the time of call. The target variable is y, records whether the client subscribed (yes) or did not (no), making this a binary classification problem.

This project has two folds first, to build a model that can help a bank prioritize which clients are worth before calling .That reduces wasted effort and cost. Secondly, to compare how different supervised learning algorithms (KNN, Decision Tree, Logistic Regression, Naive Bayes, and a Neural Network) perform on the same real-world, imbalanced dataset, also with an unsupervised K-Means analysis to see what natural groupings exist in the data independent of the target label.

# 2. Dataset Description

## Dataset Description

The given raw dataset has 21 columns: 20 input features plus the target y. After removing 12 duplicate rows, 41,176 data points remain for further work.

This is a classification problem and not a regression problem.Because the target y takes only two discrete values yes/no rather than a continuous number.

From 20 input features there are 10  numerical (age, duration, campaign, pdays, previous, emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed) and other 10 are categorical (job, marital, education, default, housing, loan, contact, month, day_of_week, outcome).

In scikit-learn models such as KNN, Logistic Regression, Naive Bayes, Decision Tree, the Neural Network and K-Means those  all require numeric input so the categorical features had to be encoded. The One-Hot Encoding (with drop_first=True to avoid redundant columns) was applied to all categorical inputs, and the target y was mapped from no/yes to 0/1.

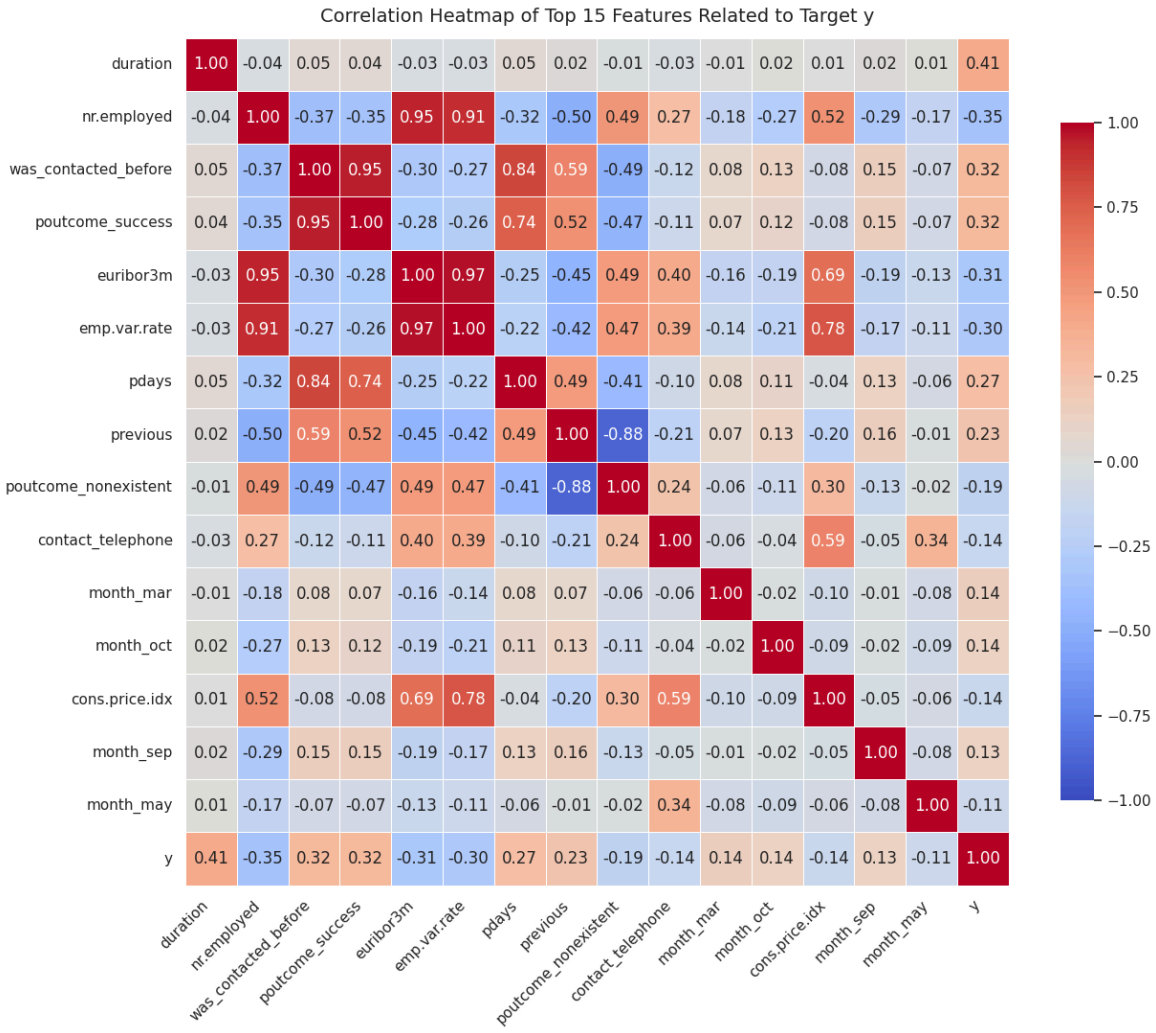



Figure 1: Full correlation heatmap of all encoded features.

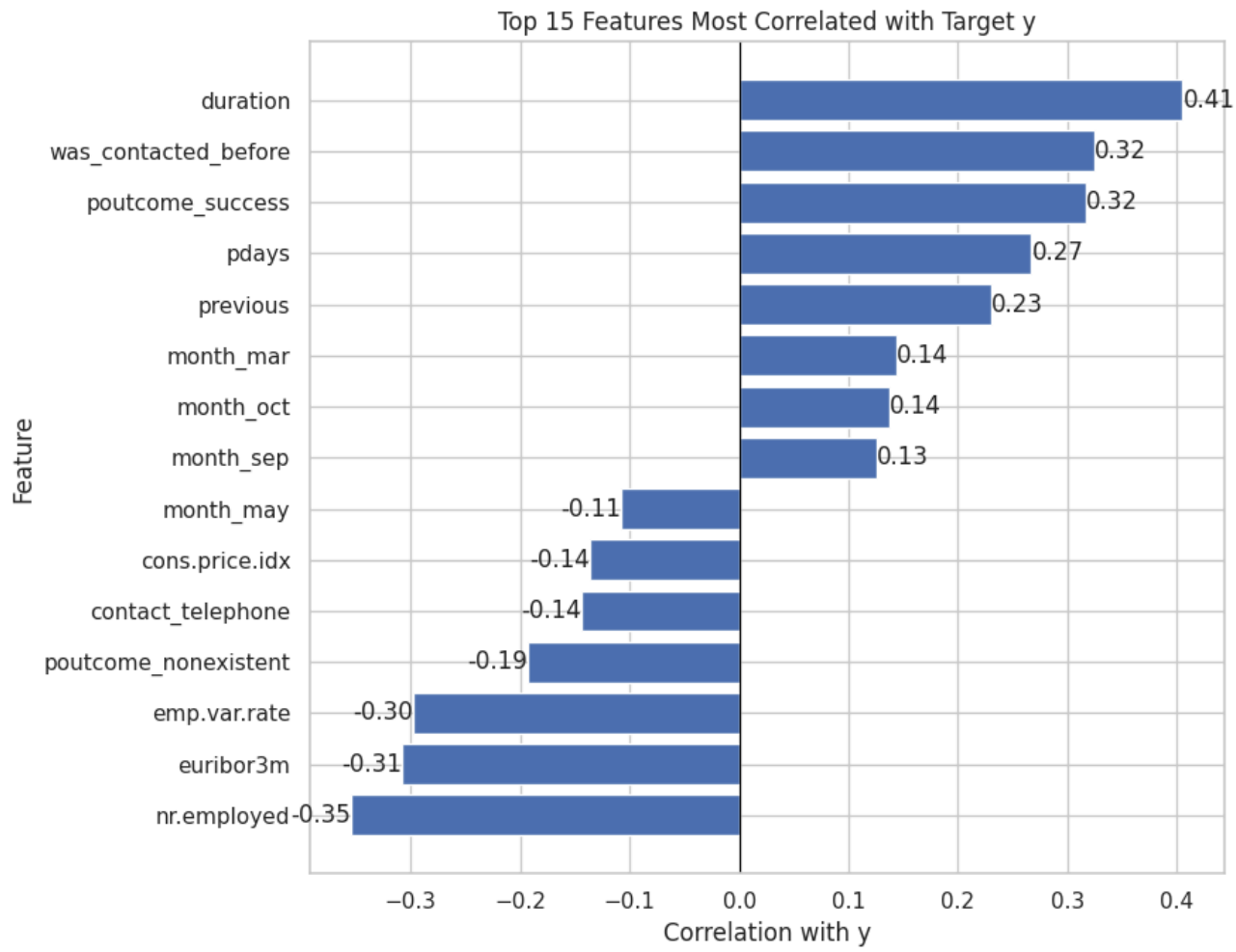



Figure 2: Top-15 feature correlation with target variable y.

What we see from the correlation analysis: mainly duration has the strongest correlation with y among all other features, which makes intuitive sense so that longer calls tend to happen when a client is engaged and more likely to say yes. However, all other features show only weak individual correlation with y that means no single feature is strongly predictive on its own and the models must need to combine many weak signals together. This also justified excluding duration from the final prediction models because it is only known after a call takes place and would leak future information into a pre-call prediction task.

## Imbalanced Dataset

The target class is imbalanced: 36,537 clients (88.73%) did not subscribe, while only 4,639 (11.27%) did.

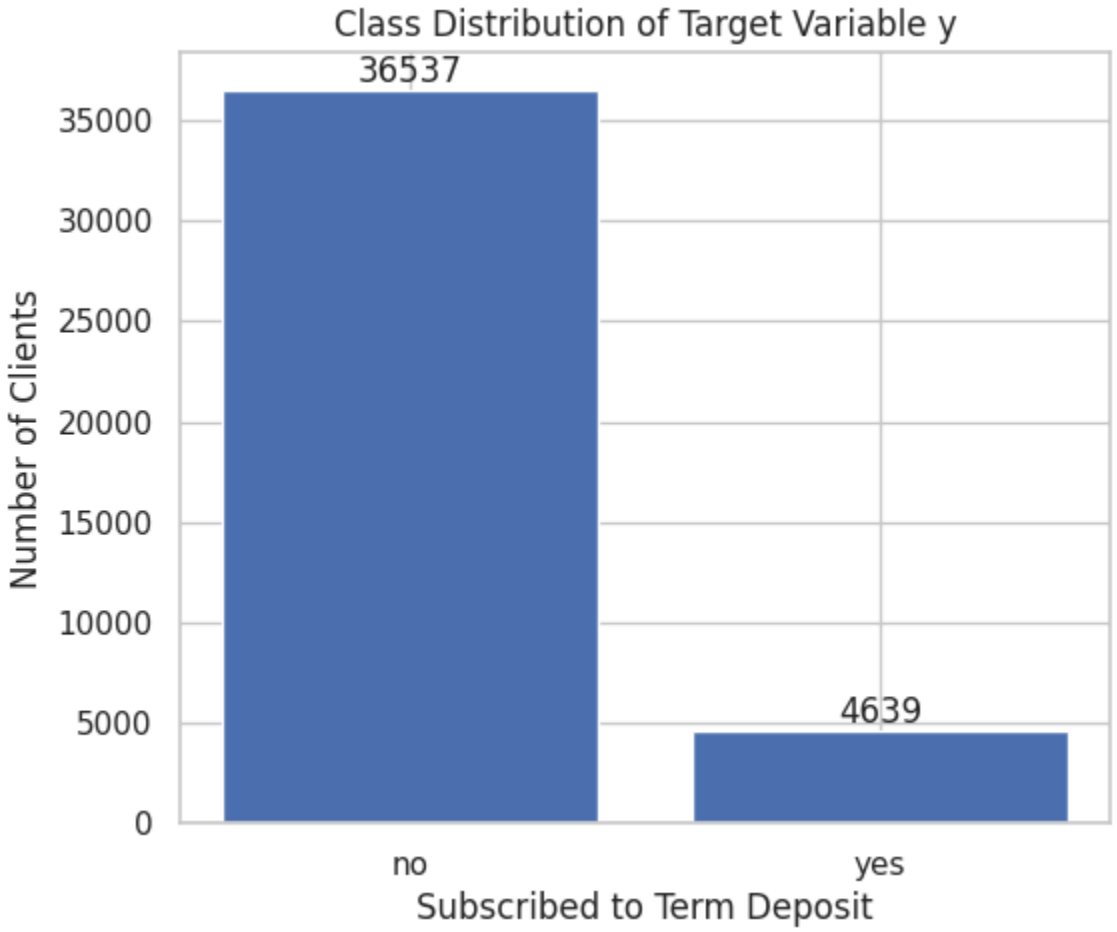



Figure 3: Class distribution of target variable y (subscribed vs. not subscribed).

This imbalance matters because a model that simply predicts 'no' for every client would already score 88.73% accuracy without learning anything useful .This majority-class baseline is used later to judge whether the trained models actually add value or not.

# Exploratory Data Analysis

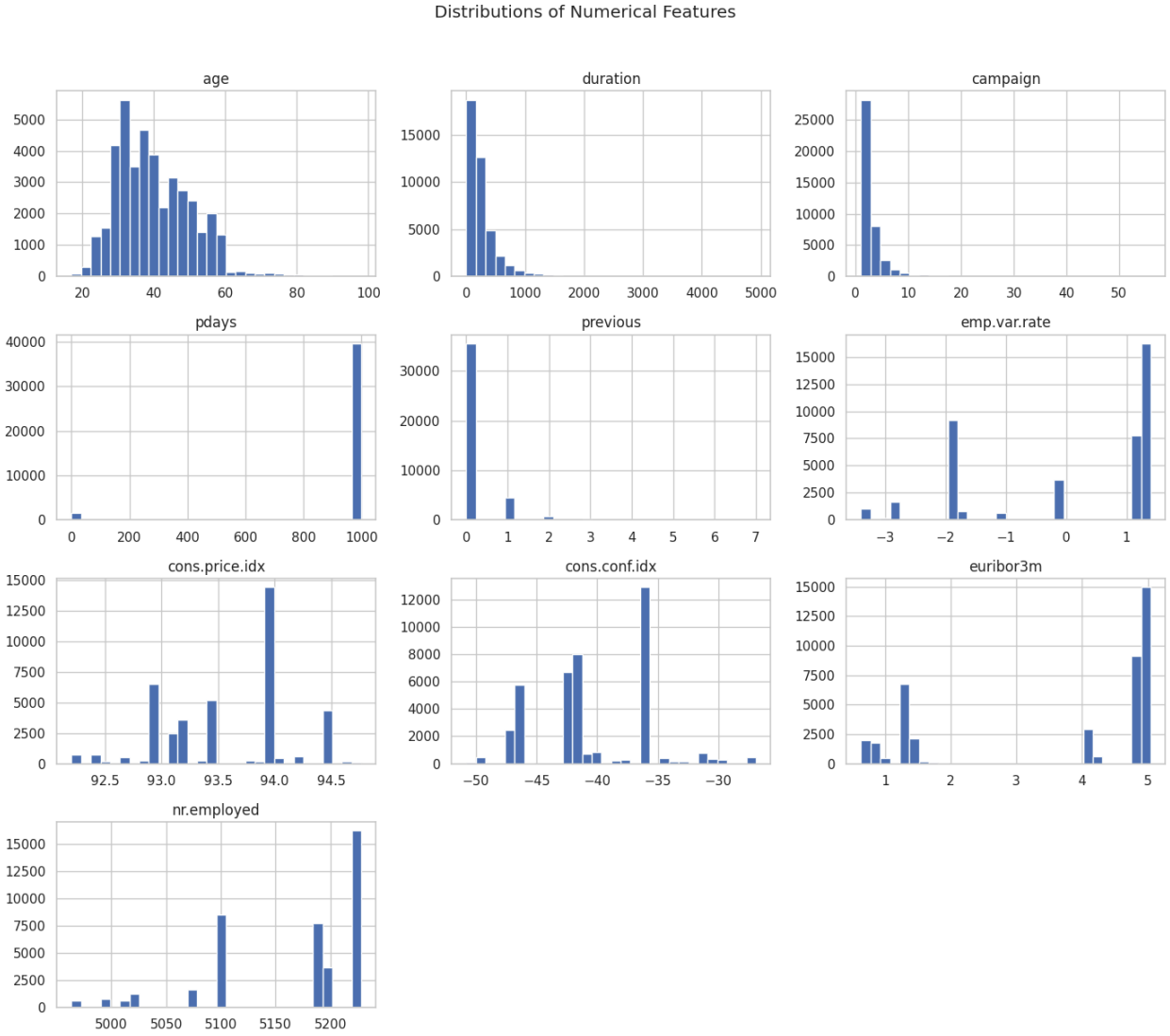



Figure 4: Histograms of numerical features.

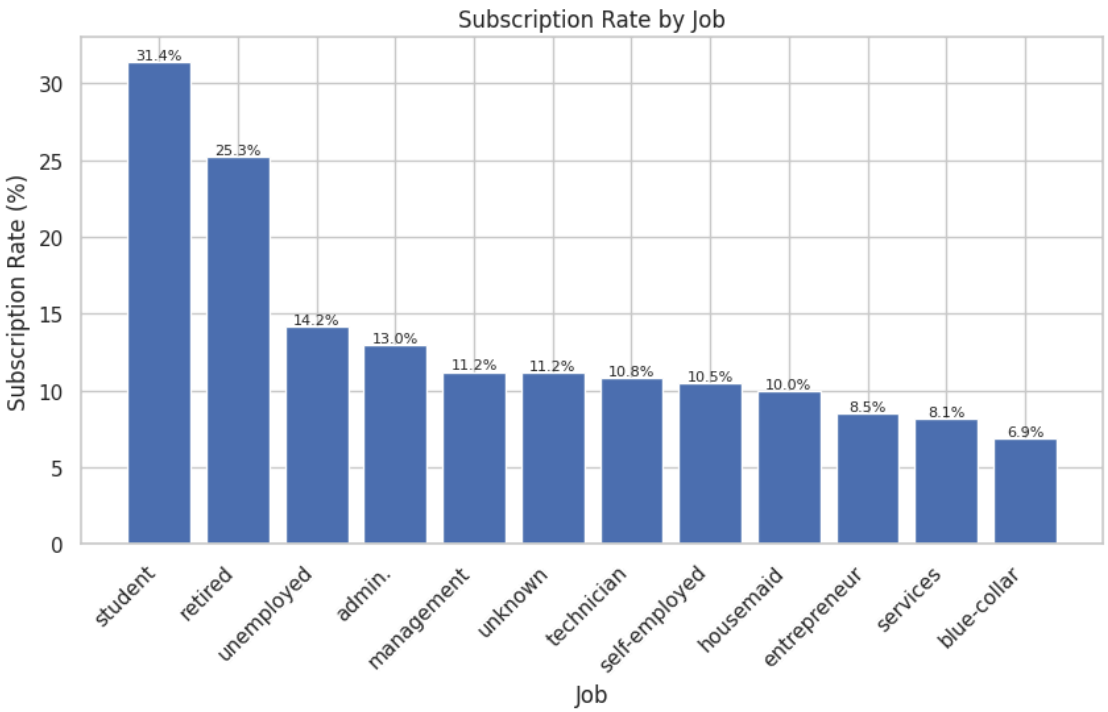



Figure 5: Subscription rate by job category.

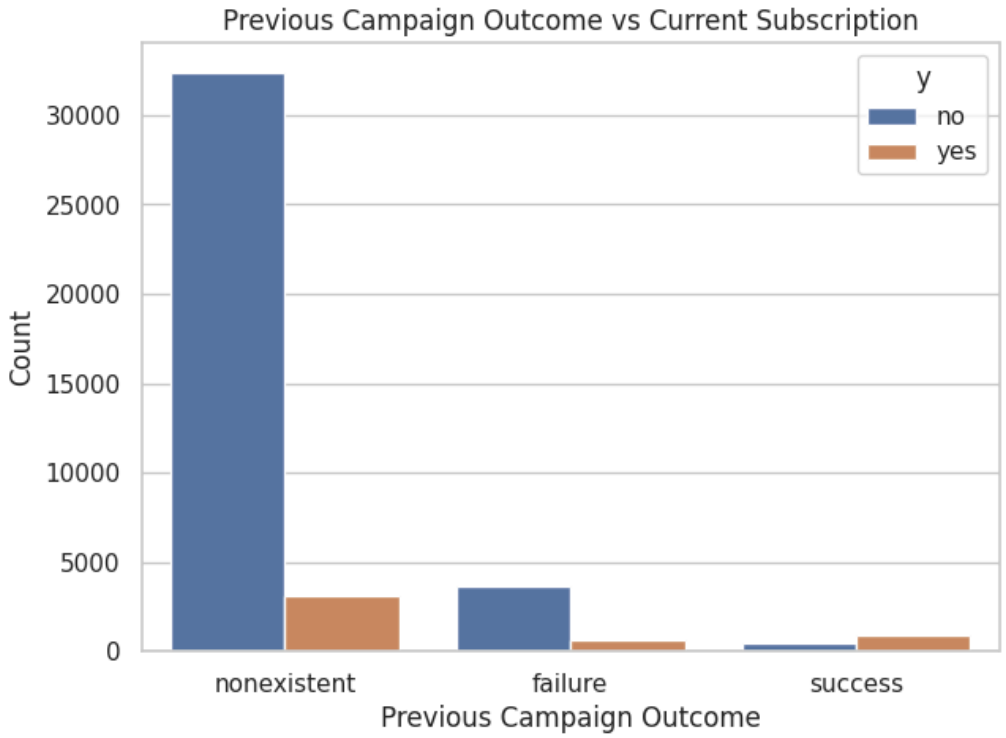



Figure 6: Previous campaign outcome vs. current subscription.

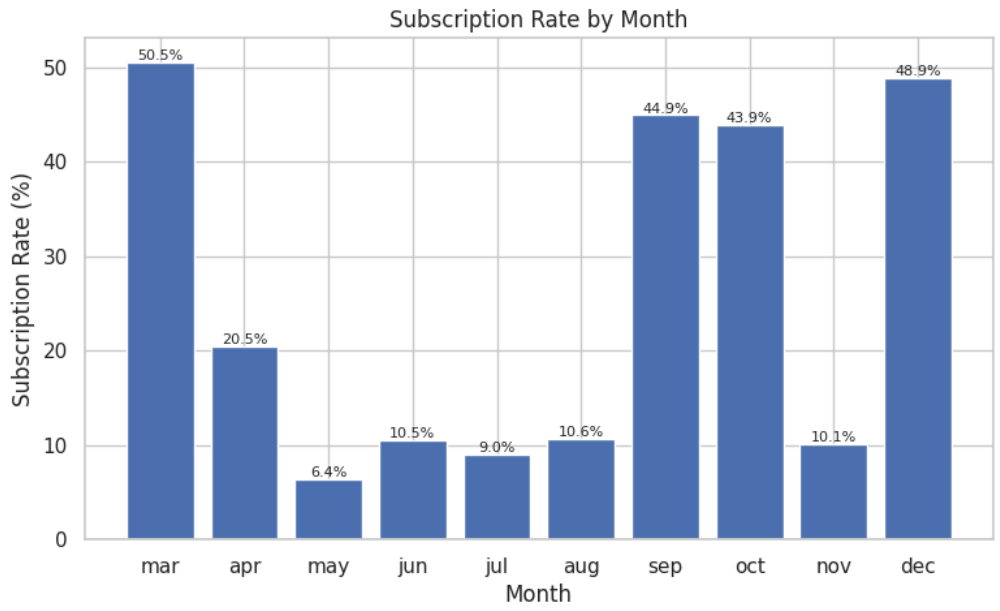



Figure 7: Subscription rate by month.

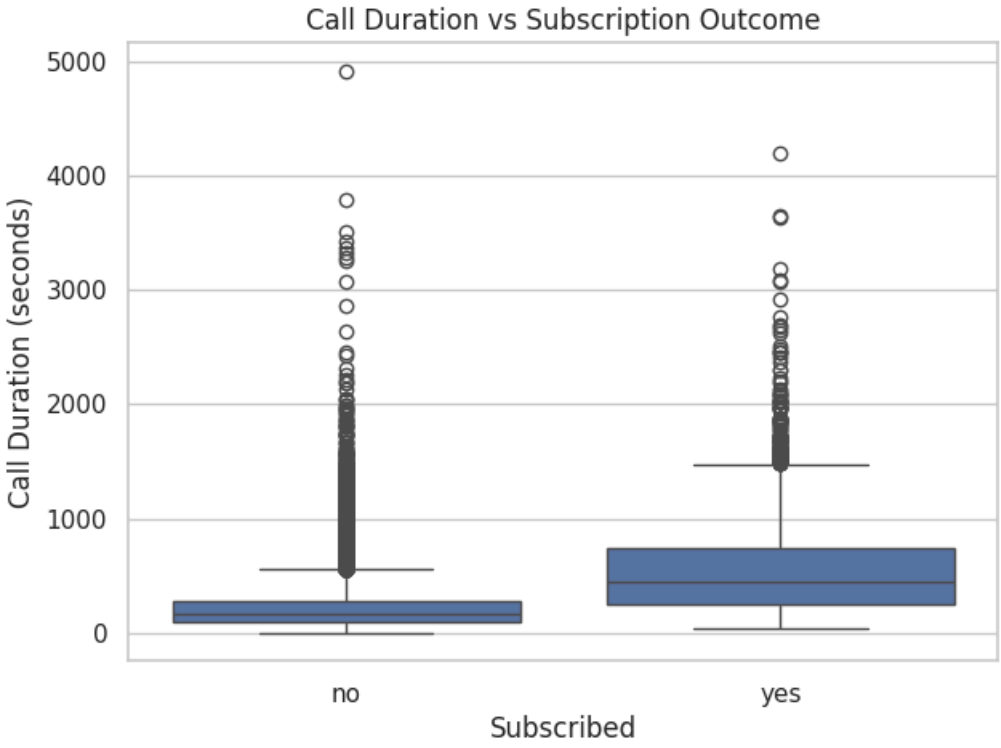



Figure 8: Call duration vs. subscription outcome (boxplot).

From the EDA, a some relationships stand out. Subscription rate mainly varies noticeably by job category with some occupations like students and retired clients showing a higher tendency to subscribe than others blue-collar workers. Those clients whose previous campaign outcome was a 'success' are far more likely to subscribe again in the current offerings compared to the clients with 'nonexistent' or 'failure' outcomes. Subscription rate also fluctuates havily by month to month basis , with some months like March, September, October, December showing higher rates than the high-volume months like May mostly reflecting different economic conditions or campaign targeting across the year. Finally, the boxplot confirms that clients who subscribed tended to have longer call durations, reinforcing why duration correlates so strongly with y but cannot be used for prediction.

# 3. Dataset Pre-processing

## Missing / Unknown values

No NaN or blank values were found anywhere in the given dataset. However, several categorical columns contain an explicit 'unknown' category: job (330 rows, 0.80%), marital (80 rows, 0.19%), education (1,730 rows, 4.20%), default (8,596 rows, 20.88%), housing (990 rows, 2.40%) and loan (990 rows, 2.40%).

Solution: 'unknown' was deliberately kept as its own valid category instead of being imputed away, since a client's refusal or inability to answer may itself carry predictive meaning for example, high 'unknown' in default could reflect privacy-conscious or higher-risk clients. As a general safety net, a basic standard imputation rule was still defined so numeric columns would be filled with the median and categorical columns with the mode but it was not actually triggered here since no true missing values existed.

## Placeholder value in pdays

The pdays column uses 999 as a sentinel value meaning 'this client was never contacted in a previous campaign', not a literal count of 999 days. Leaving this as-is would make the feature look like an extreme outlier and distort any distance-based or scaled model.

Solution: a new binary feature, was_contacted_before, was created 1 if pdays != 999, else 0 and then the placeholder 999 was replaced with 0 in pdays itself. This way the 'never contacted' information is preserved explicitly rather than being encoded as a misleading large number. After this transformation, 39,661 clients were flagged as not previously contacted and 1,515 as previously contacted.

## Categorical values

Text-based categorical columns such as job, marital, education cannot be used directly by distance-based or numeric models like KNN, Logistic Regression, the Neural Network, Naive Bayes, Decision Tree, or K-Means.

Solution: One-Hot Encoding (drop_first=True) was applied to all categorical input columns and the target y was mapped to 0 (no) and 1 (yes). After encoding, the final feature set contains 53 columns.

## Feature scaling

The numerical features span very different ranges such as age ranges roughly 17–98 while nr.employed ranges in the thousands. Without scaling, features with larger numeric ranges would dominate distance-based and gradient-based algorithms.

Solution: StandardScaler was applied to the numeric features for KNN, Logistic Regression, the Neural Network, and K-Means (all distance/gradient-sensitive methods), while Decision Tree and Naive Bayes were trained on the unscaled encoded values since tree-based and probability-based methods do not require scaling.

### Data leakage from duration

The duration feature length of the marketing call in seconds is only known after the call has already happened. So using it as a predictive input would let the model 'cheat' by seeing information that would not actually be available at prediction time in a real deployment.

Solution: so the duration was retained for EDA and correlation analysis (to understand the data) but it was excluded from the input features used to train every supervised model, ensuring the models reflect a realistic pre-call prediction scenario.

# 4. Dataset Splitting

Because the target class is imbalanced (11.27% positive), a stratified split was used throughout so that the proportion of yes/no clients stays consistent across all the subsets.

The data was first split 70% train / 10% validation / 20% test as standard procedure. The validation set was used to select simple hyperparameters for KNN (number of neighbors K and weighting scheme), Decision Tree (max depth and minimum samples per leaf), and Logistic Regression (regularization strength C) by comparing validation accuracy across a small manually-chosen grid of values, rather than using an automated method like GridSearchCV. This kept the tuning process very simple and easy to explain.

After selecting the best hyperparameters the training and validation sets were combined to form a final 80% training set and each model was retrained on this combined set and evaluated exactly once on the untouched 20% test set so the test set was never used during hyperparameter selection, which avoids optimistic bias in the reported results and that is important.

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| Split | Size | Positive class (%) |
|---|---|---|
| Final Train (80%) | 32,940 | 11.27% |
| Final Test (20%) | 8,236 | 11.27% |

# 5. Model Training & Testing

## Supervised Models

Five supervised models were trained on the final 80% training set and evaluated on the same held-out 20% test set:

- KNN — best hyperparameters found: K=31, uniform weighting (validation accuracy 0.9024)
- Decision Tree — best hyperparameters found: max_depth=4, min_samples_leaf=10 (validation accuracy 0.9014)
- Logistic Regression — best hyperparameter found: C=0.01 (validation accuracy 0.9014)
- Naive Bayes — Gaussian Naive Bayes, no hyperparameter tuning needed
- Neural Network — Keras Sequential model: Dense(64, ReLU) → Dense(32, ReLU) → Dense(1, sigmoid), Adam optimizer, binary cross-entropy loss, trained with EarlyStopping (patience=3, monitor=val_loss); training stopped after 12 epochs, restoring weights from epoch 9

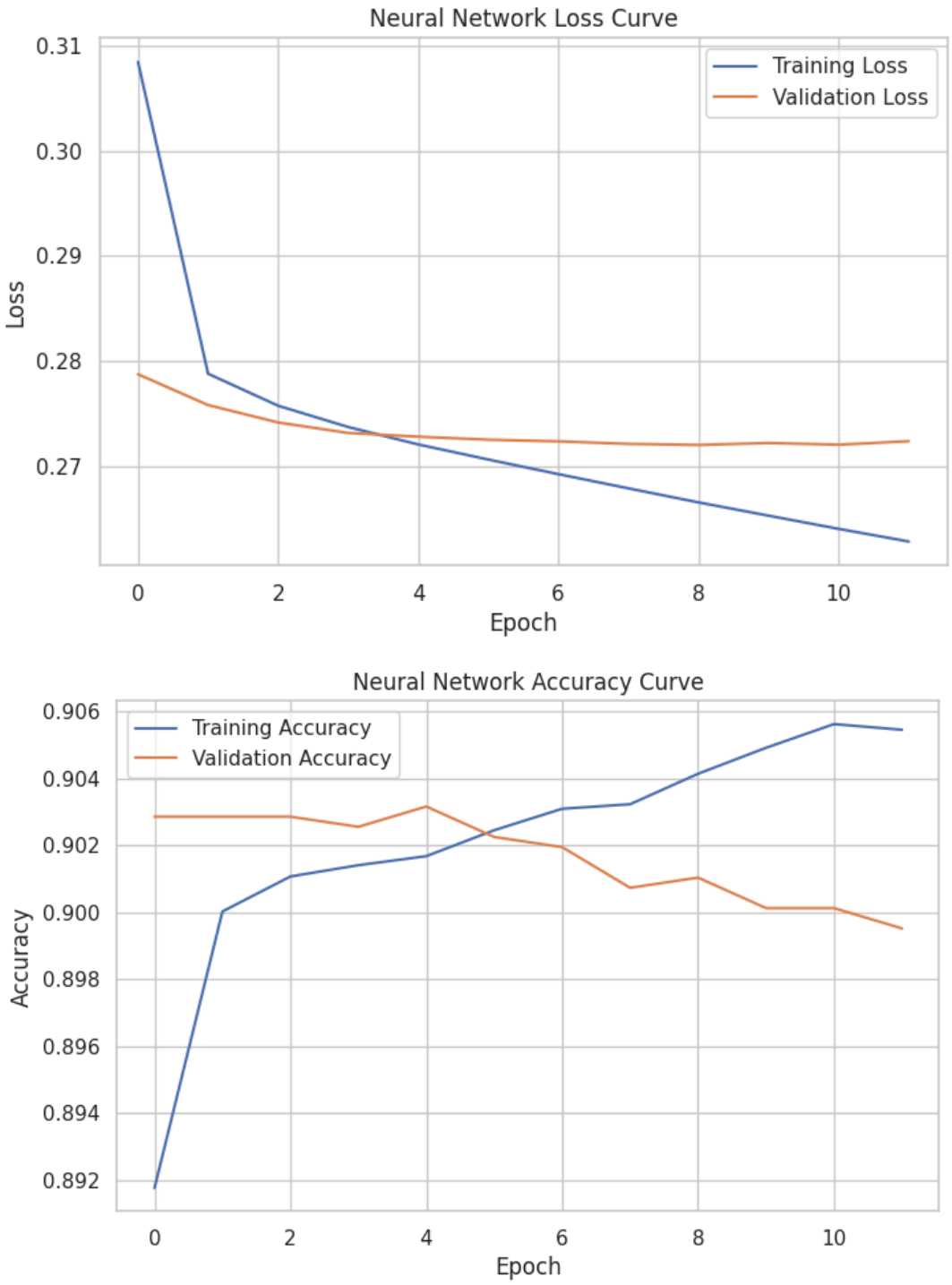



Figure 9: Neural Network training loss and accuracy curves.

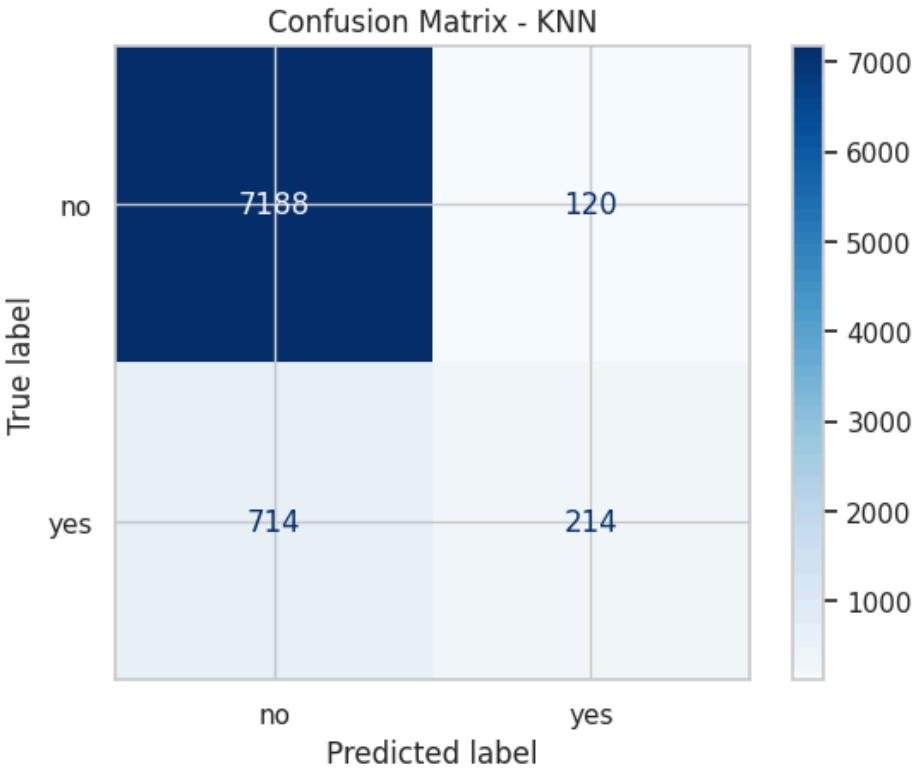



Figure 10: Confusion matrix - KNN.

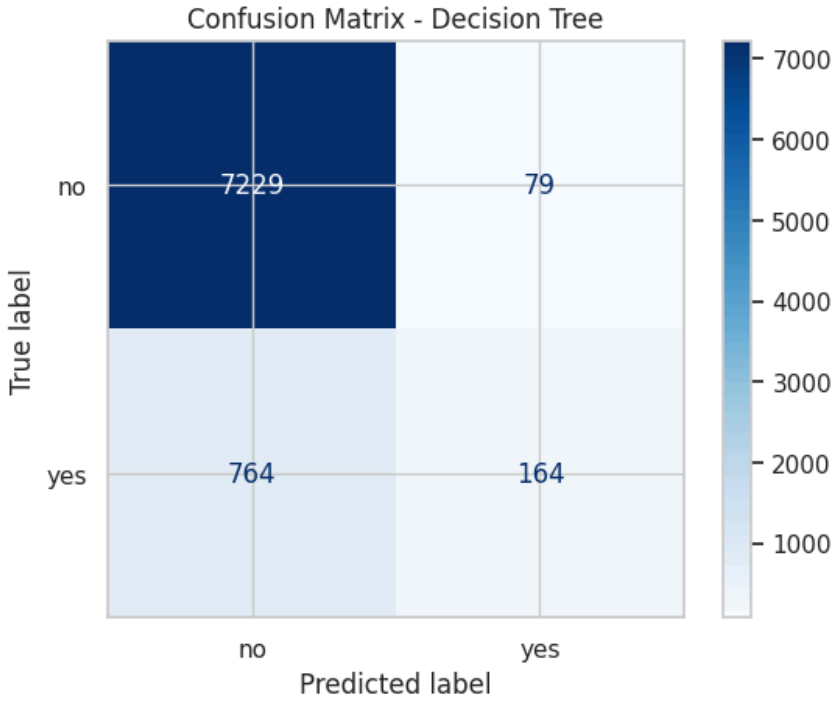



Figure 11: Confusion matrix - Decision Tree.

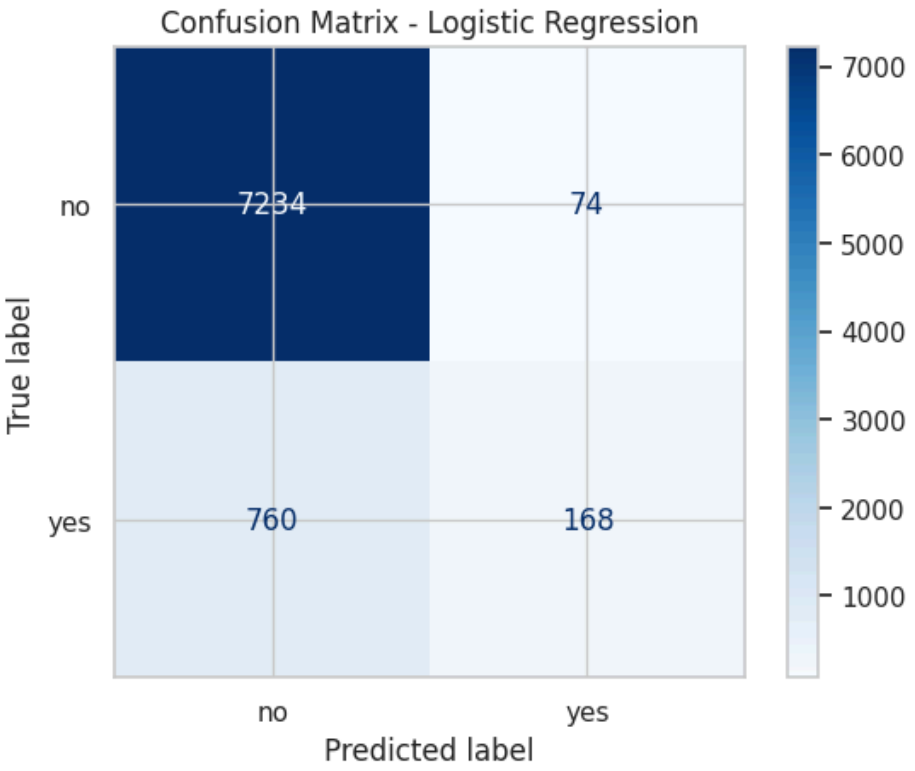



Figure 12: Confusion matrix - Logistic Regression.

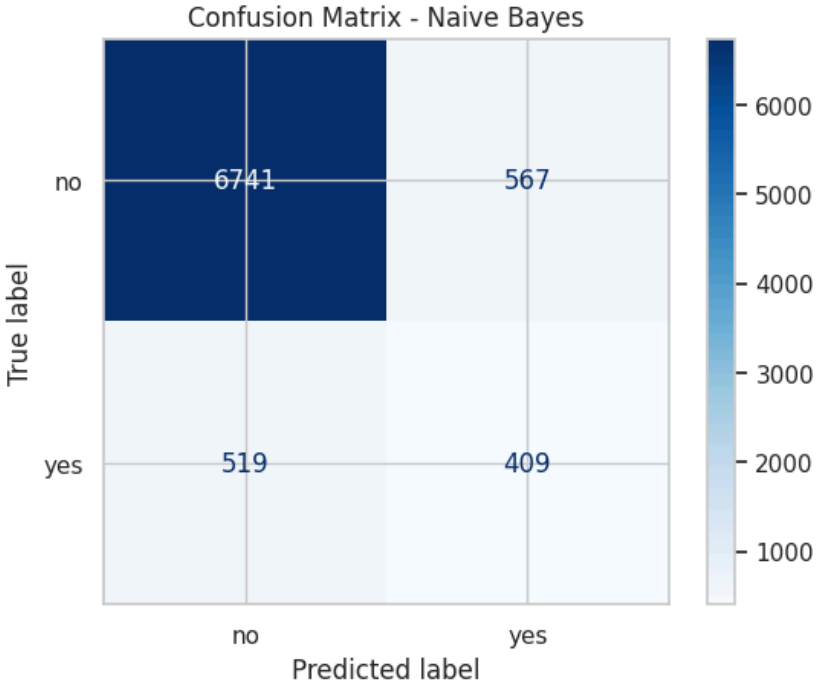



Figure 13: Confusion matrix - Naive Bayes.

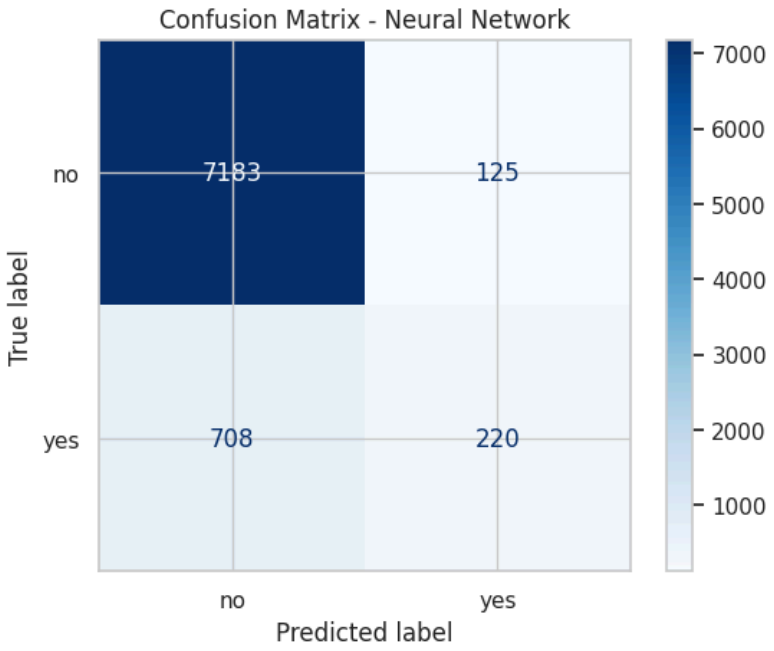



Figure 14: Confusion matrix - Neural Network.

## Unsupervised Learning — K-Means

K-Means clustering was applied to the same scaled 80% training feature set without using the y label at all, to see what natural groupings exist among clients based purely on feature similarity.

To choose the number of clusters K, both the Elbow Method (Sum of Squared Errors) and the Silhouette Score were computed for K = 2 through 8. The silhouette score was highest at K=2 (0.3131), so K=2 was selected as the final number of clusters.

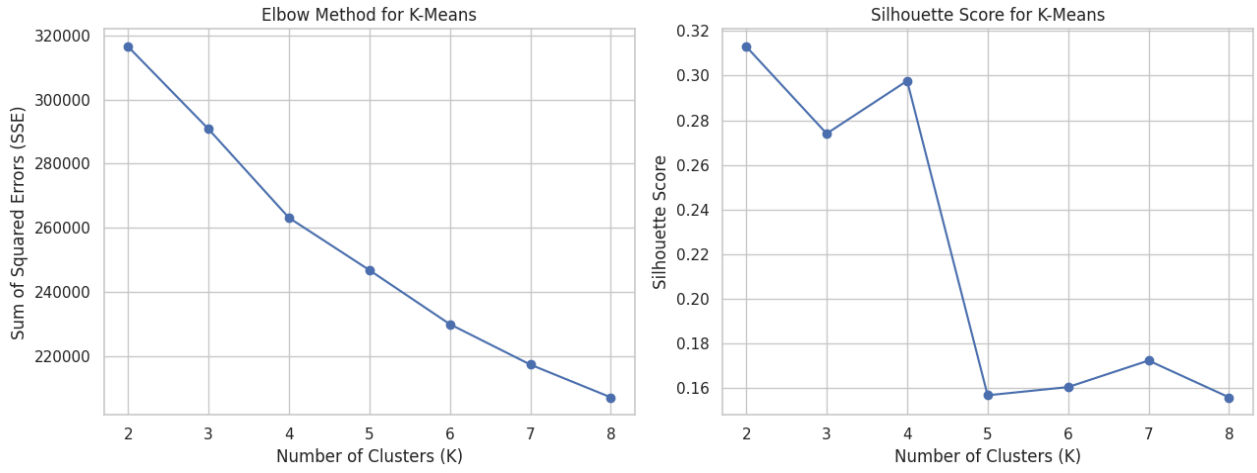



Figure 15: Elbow method (SSE) and Silhouette Score plots for K-Means.

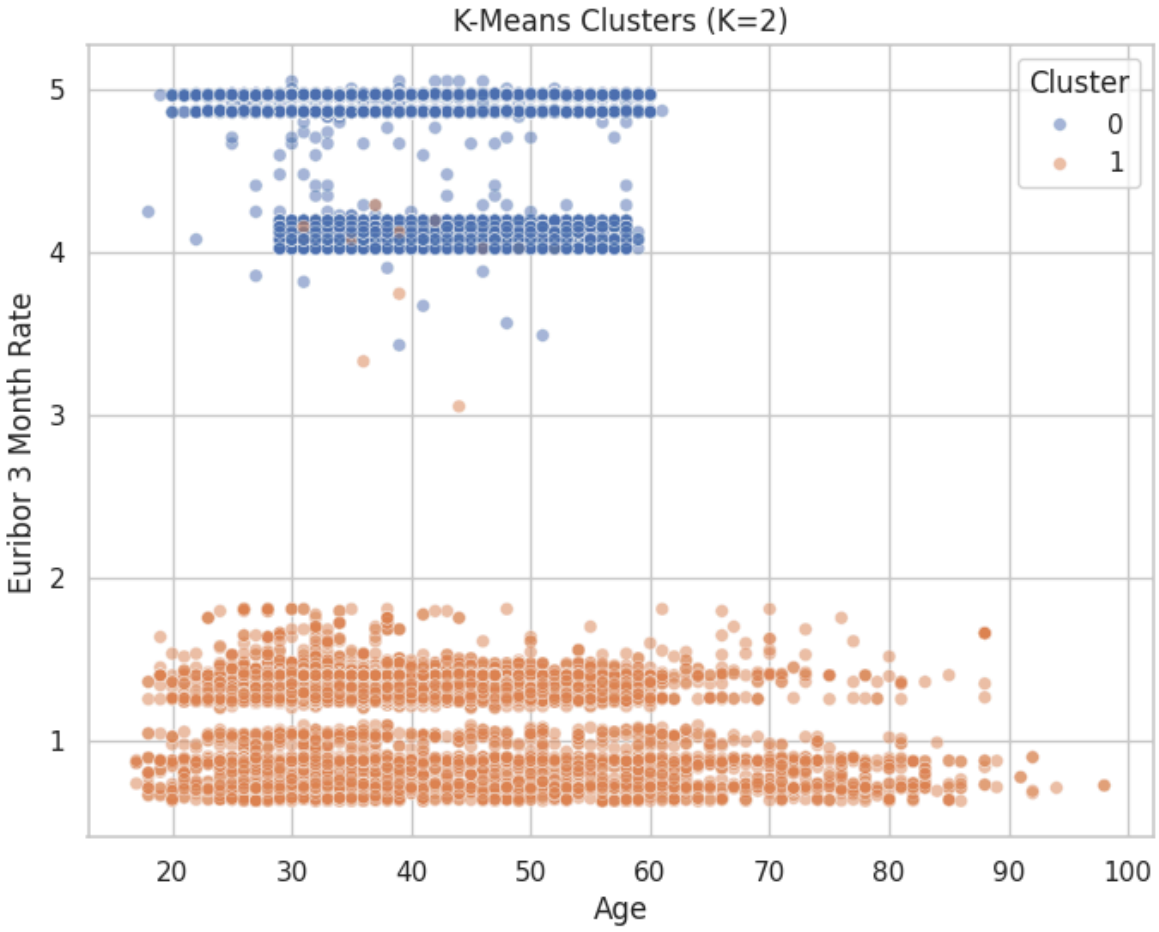



Figure 16: K-Means cluster scatter plot (age vs. euribor3m).

The final clustering produced Cluster 0 with 22,078 members and Cluster 1 with 10,862 members. When compared against the actual y labels (which K-Means never saw), Cluster 0 contained 20,991 'no' and 1,087 'yes' clients, while Cluster 1 contained 8,238 'no' and 2,624 'yes' clients. This shows Cluster 1 has a noticeably higher concentration of subscribers relative to its size, suggesting K-Means is picking up on a real underlying structure in the economic/demographic features that correlates with subscription likelihood — even though K-Means clusters do not need to, and did not, align perfectly with the yes/no labels, since it groups purely by feature similarity, not by the target.

# 6. Model Selection / Comparison Analysis

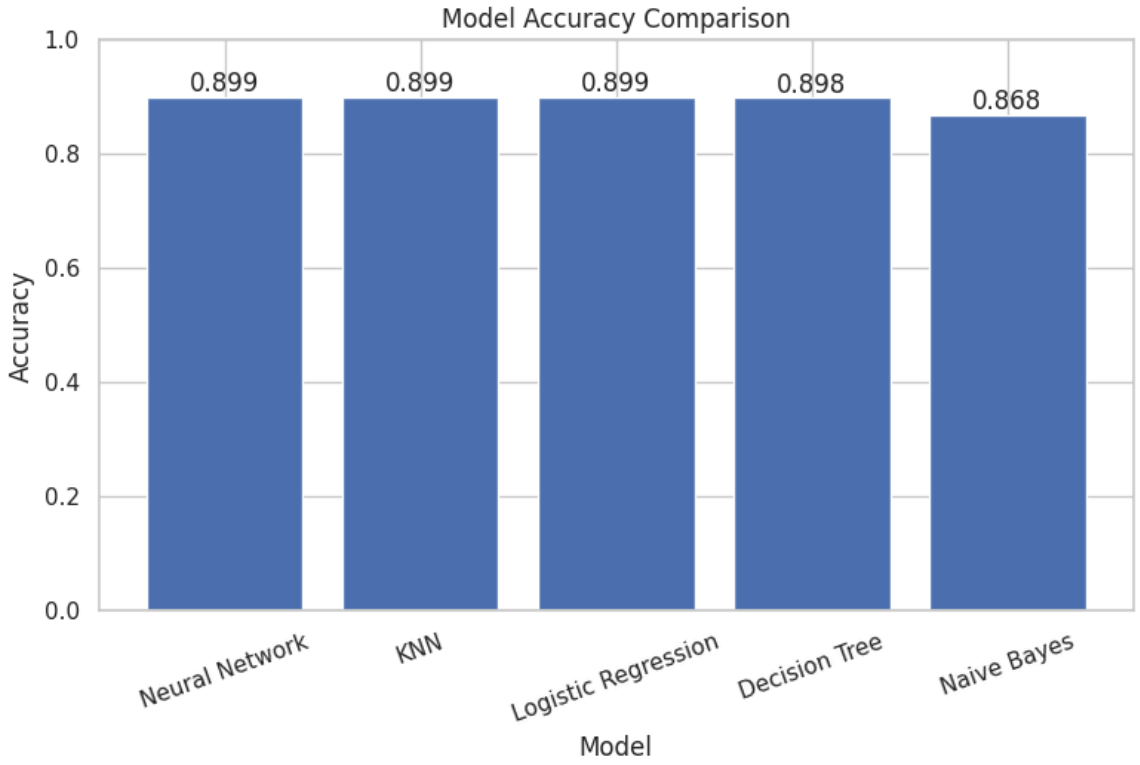



Figure 17: Accuracy comparison across all models.

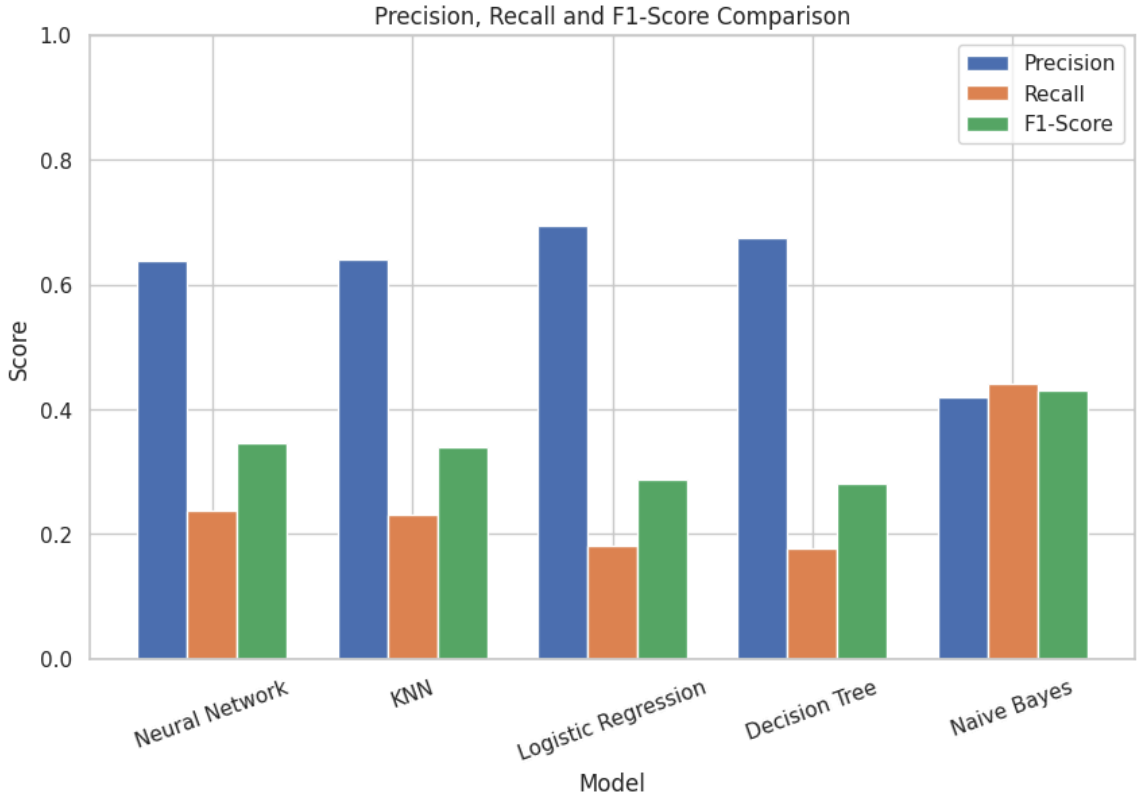



Figure 18: Precision, Recall, and F1-Score comparison across all models.

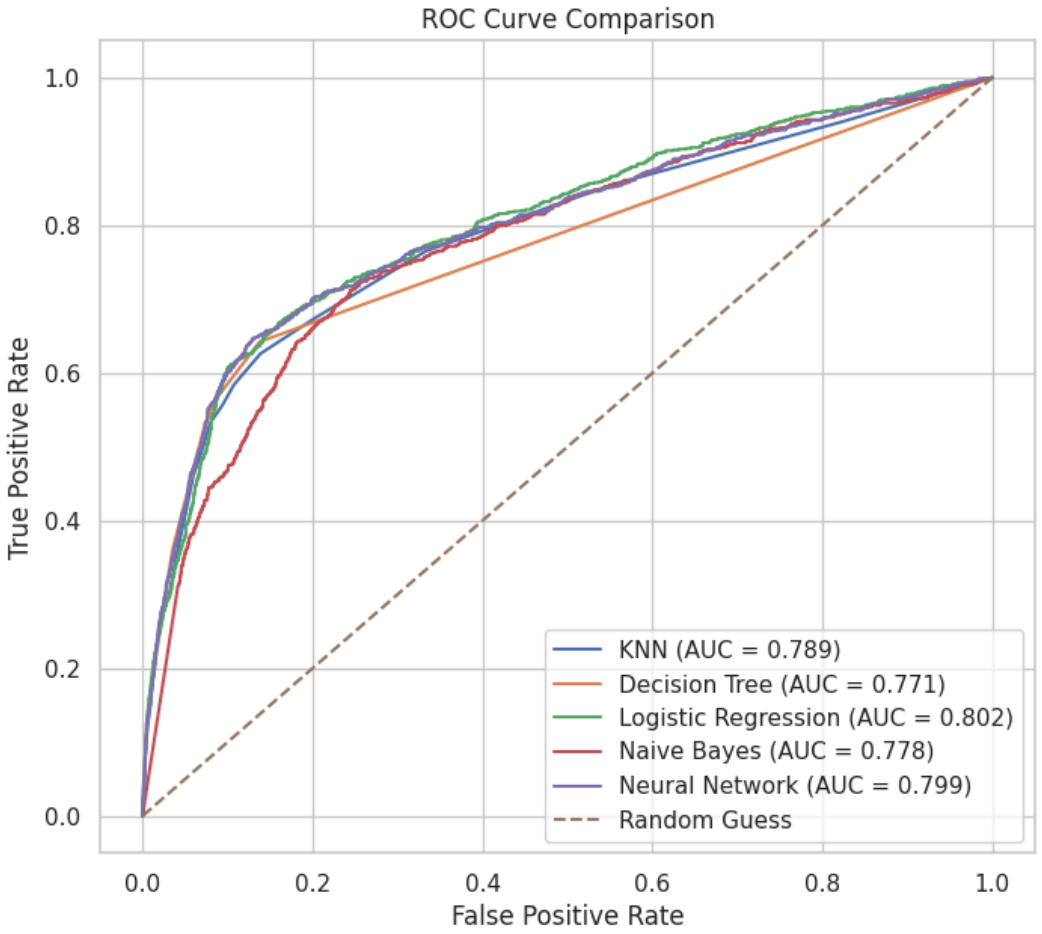



Figure 19: ROC curves and AUC scores for all five models.

All five models were evaluated on the identical 20% test set (8,236 clients). The majority-class baseline accuracy — i.e. always predicting 'no' — is 0.8873.

| Model | Accuracy | Precision | Recall | F1-Score | AUC |
|---|---|---|---|---|---|
| Neural Network | 0.8989 | 0.6377 | 0.2371 | 0.3456 | 0.7991 |
| KNN | 0.8987 | 0.6407 | 0.2306 | 0.3391 | 0.7890 |
| Logistic Regression | 0.8987 | 0.6942 | 0.1810 | 0.2872 | 0.8021 |
| Decision Tree | 0.8976 | 0.6749 | 0.1767 | 0.2801 | 0.7709 |
| Naive Bayes | 0.8681 | 0.4191 | 0.4407 | 0.4296 | 0.7779 |

Highest test accuracy: Neural Network (0.8989), closely followed by KNN and Logistic Regression  all four discriminative models cluster tightly around 89–90% accuracy, only slightly above the 88.73% majority baseline.

Highest AUC: Logistic Regression (0.8021), narrowly ahead of the Neural Network (0.7991), indicating Logistic Regression ranks positive vs negative clients slightly better across all classification thresholds even though its raw accuracy is not the highest.

Highest Precision: Logistic Regression (0.6942) — when it predicts 'yes', it is correct about 69% of the time, the most reliable of the five models on this measure.

Highest Recall (minority class): Naive Bayes (0.4407) — it catches the largest share of actual subscribers, though at the cost of more false positives, reflected in its lower precision and accuracy.

Highest F1-Score: Naive Bayes (0.4296) — because it balances precision and recall best for the minority 'yes' class, even though its overall accuracy (0.8681) is the lowest of the five models.

Overall, no single model wins on every metric, which is expected on an imbalanced dataset. These models that optimize for overall accuracy (Neural Network, KNN, Logistic Regression, Decision Tree) tend to under-predict the rare 'yes' class, while Naive Bayes trades some accuracy for much better minority-class recall and F1-score. This is why accuracy alone is a misleading metric here  a trivial model that always predicts 'no' already reaches 88.73% accuracy, so the real value of these models has to be judged through precision, recall, F1-score, and AUC together, not accuracy in isolation.

# 7. Conclusion

This project created and evaluated five supervised learning models – KNN, Decision Tree, Logistic Regression, Naive Bayes, and a Neural Network – to determine if a bank client would take out a term deposit and an unsupervised K-Means analysis to examine natural groupings of clients.

All four discriminative models (Neural Network, KNN, Logistic Regression, Decision Tree) have similar performance in terms of accuracy (0.898-0.899), marginally outperforming the baseline majority class accuracy of 0.8873; Naive Bayes has a higher F1 score and recall on the minority class (0.726 and 0.884, respectively), at the cost of lower overall accuracy. It is a natural result of the class imbalance (88.73% no class vs 11.27% yes class) that models optimised for accuracy will naturally favour the more prevalent class; hence, precision, recall, F1-score and AUC measures had to be analysed in conjunction with accuracy rather than solely by this metric.

duration was intentionally omitted from all predictive models as it is the most highly associated feature with y, and is not available until after the call has completed, making it data-leaking in a true pre-call prediction scenario. The models used the unscaled encoded data for categorical features (Job, Marital, Education, etc.) and Standard Scaled data for the numeric features for the distance/gradient based models (KNN, Logistic Regression, Neural Network, K-Means).The distance/gradient based models (KNN, Logistic Regression, Neural Network, K-Means) used Standard Scaled data for numeric features (Age, Duration, Job, Marital, etc.), while the tree and probability based models (Decision Tree, Naive Bayes) used the unscaled encoded data for the same features. There were no true missing values, however several columns contained an 'unknown' class which was not imputed but rather considered as its own class as it may contain an actual signal. The value pdays = 999 was special-cased by converting to a new binary was_contacted_before and replacing 999 with 0 to prevent it from being interpreted as an extreme numeric-outlier.

In the K-Means analysis, without knowledge of the target label, two natural clusters (K=2, silhouette score 0.3131) were identified. One cluster had a visibly greater proportion of actual subscribers than the other, indicating that there is structure in the actual features that does relate to the label — though the K-Means clusters are neither expected nor necessarily desired to be perfectly aligned with the labels, as they are clustered by how similar the features are to each other, not by the label itself.

The primary difficulties in this project included dealing with the high class imbalance, where accuracy was not a reliable measure on its own, since it risked overfitting; transforming a large number of categorical variables into actionable numeric data without adding too many features; understanding and addressing the pdays=999 "placeholder" to prevent it from dominating the numeric distribution; and ensuring that the duration was not fed back into the model, as it was highly correlated with the result. It was possible to compare all five models fairly, and realistically, as a result of addressing these issues in the careful preprocessing, stratified splitting and the hyperparameter search using a validation set.